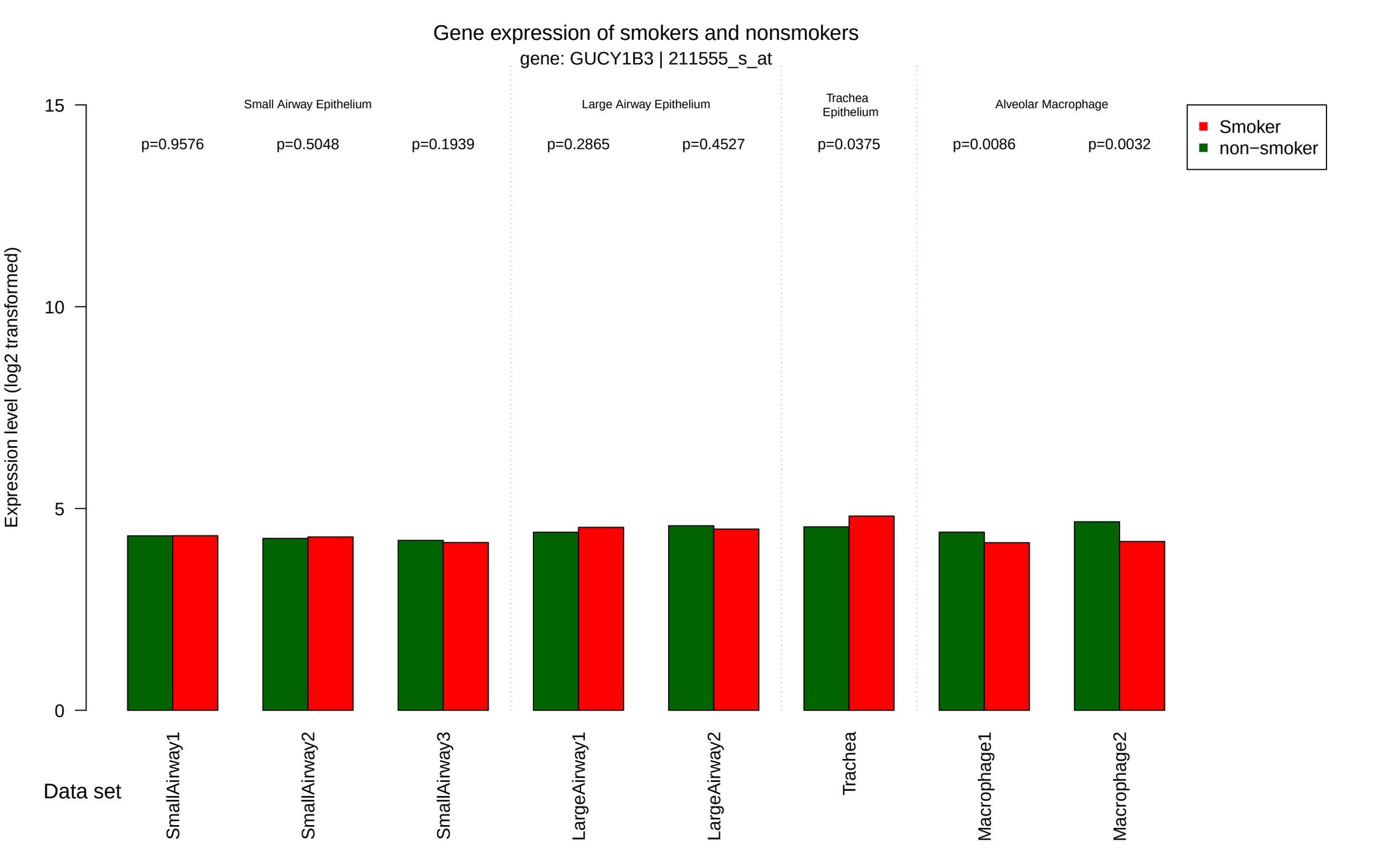
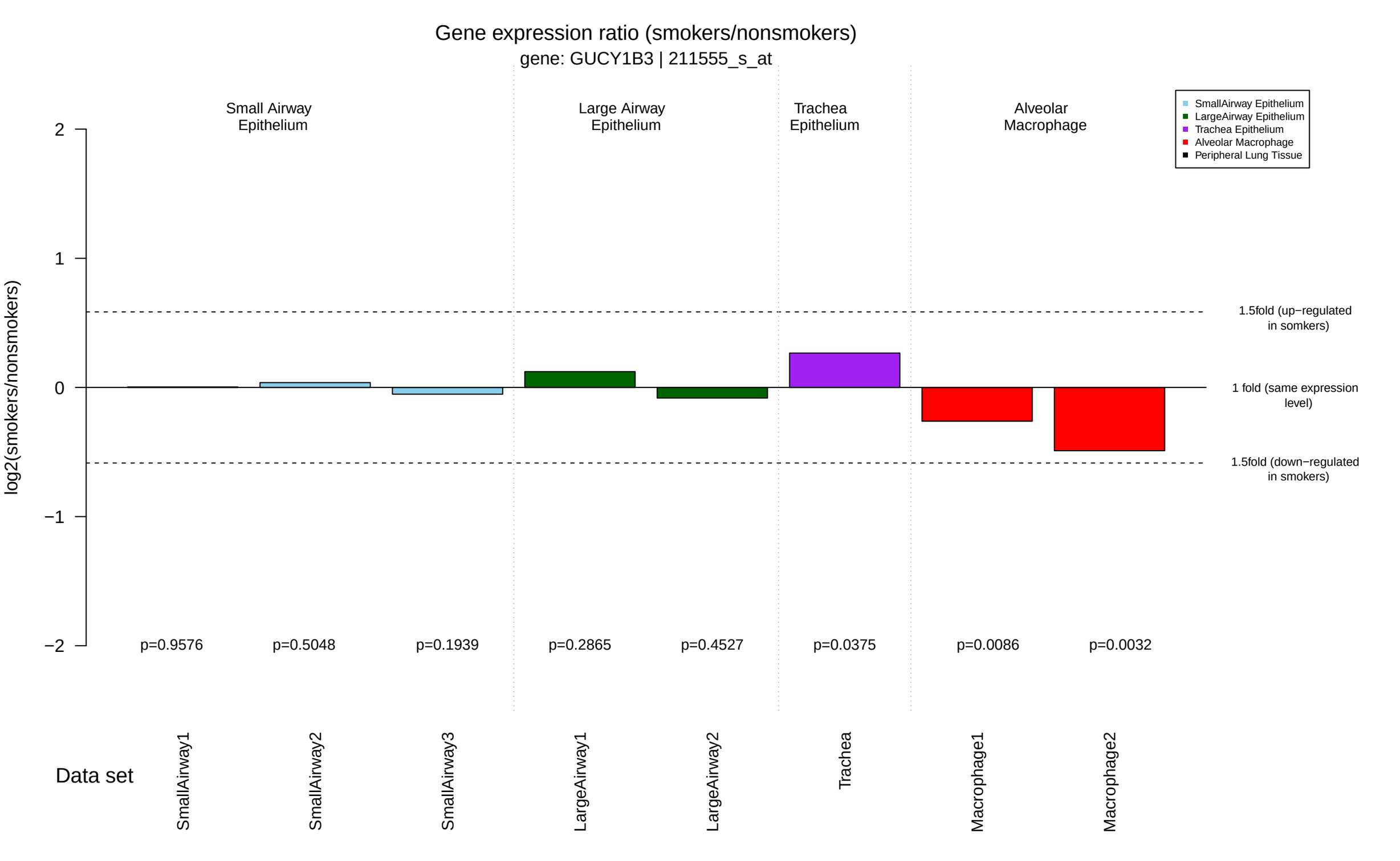
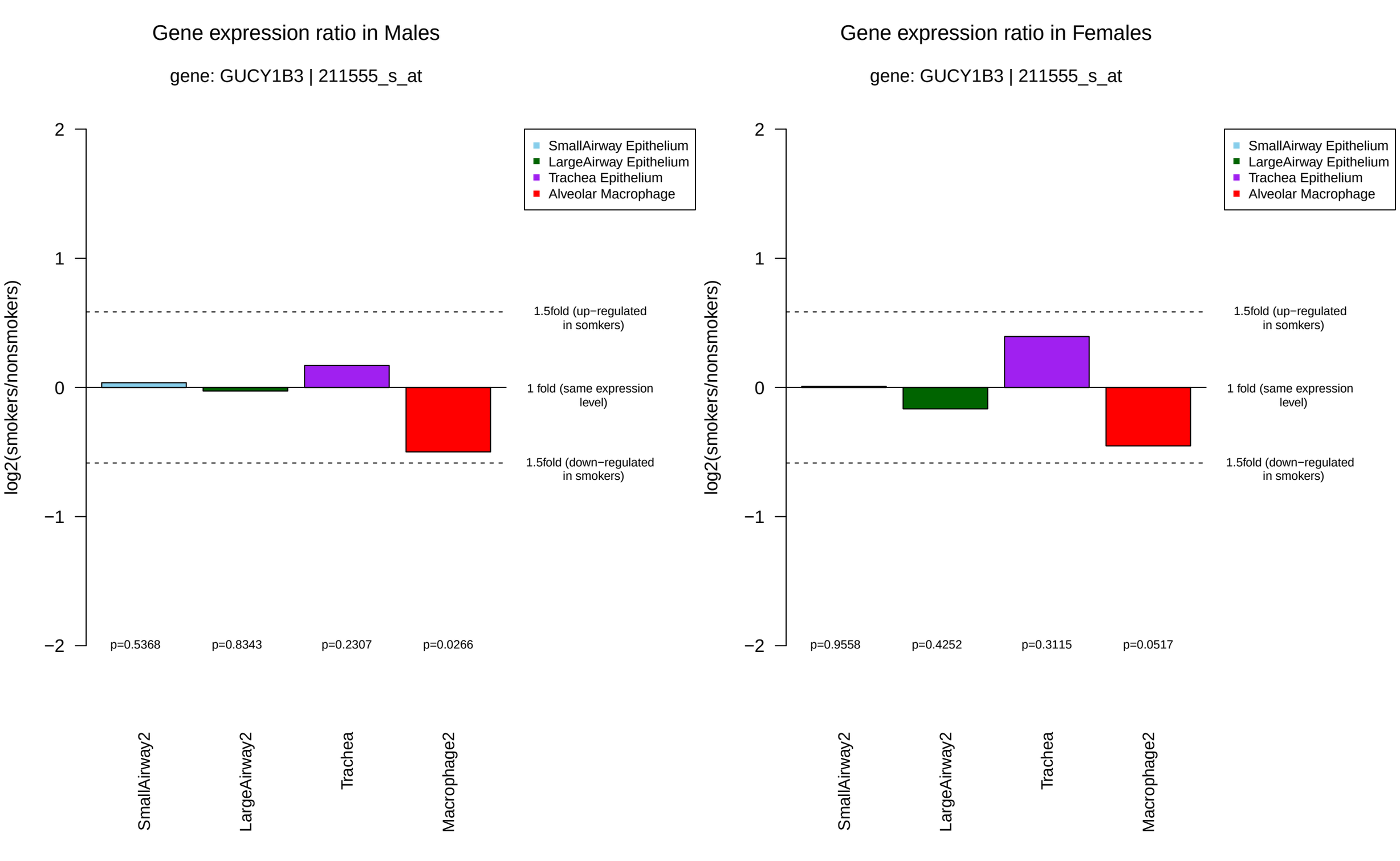
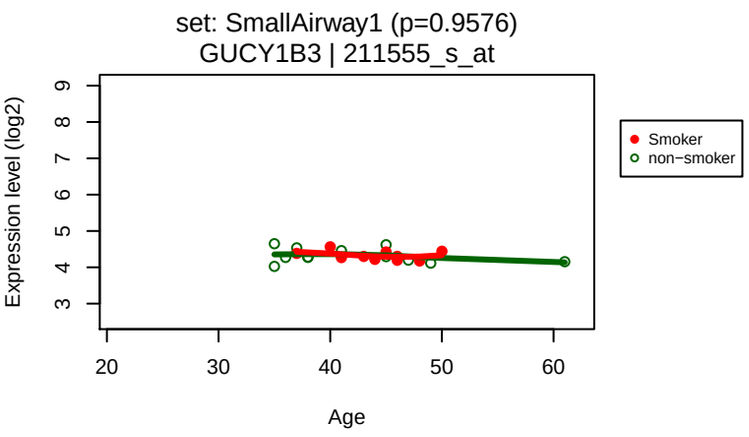
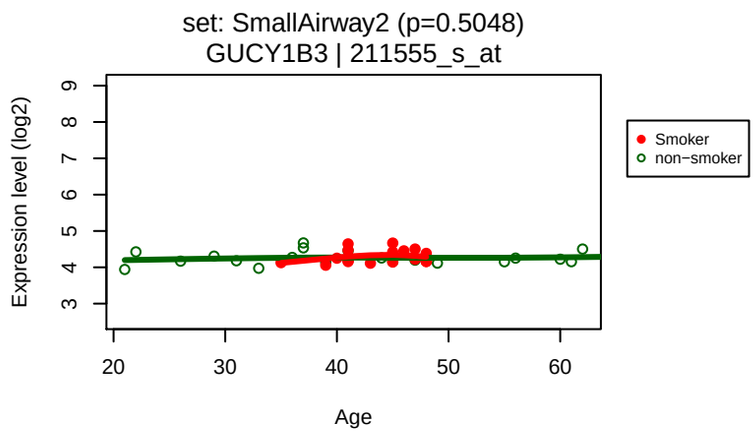
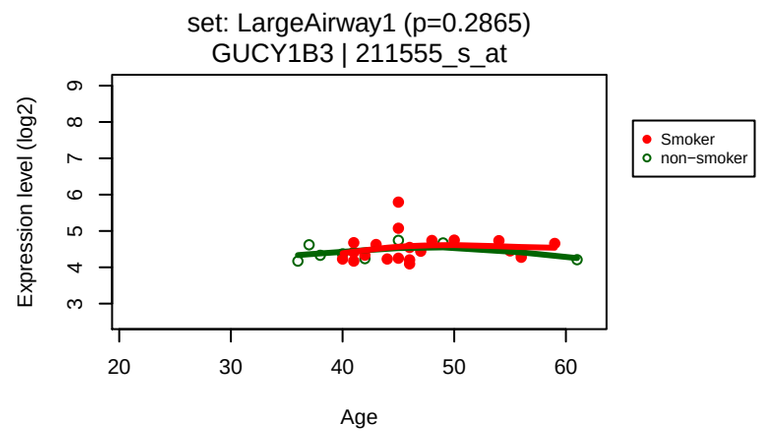

Gene expression of smokers and nonsmokers
gene: GUCY1B3 | 211555_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9576
p=0.5048
p=0.1939
p=0.2865
p=0.4527
p=0.0375
p=0.0086
p=0.0032
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

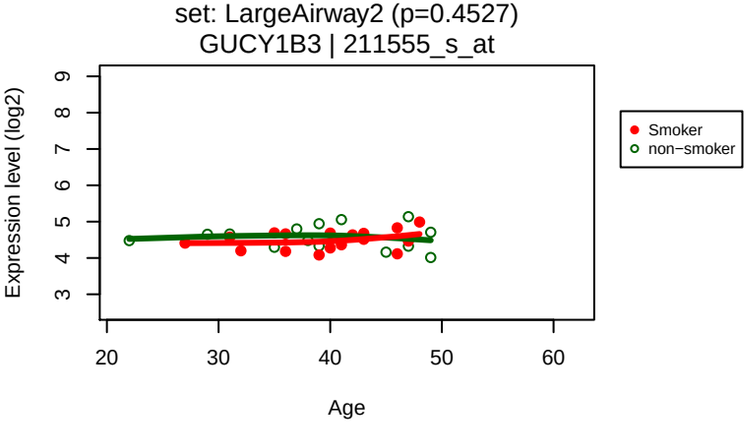

Gene expression ratio (smokers/nonsmokers)
gene: GUCY1B3 | 211555_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9576
p=0.5048
p=0.1939
p=0.2865
p=0.4527
p=0.0375
p=0.0086
p=0.0032
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

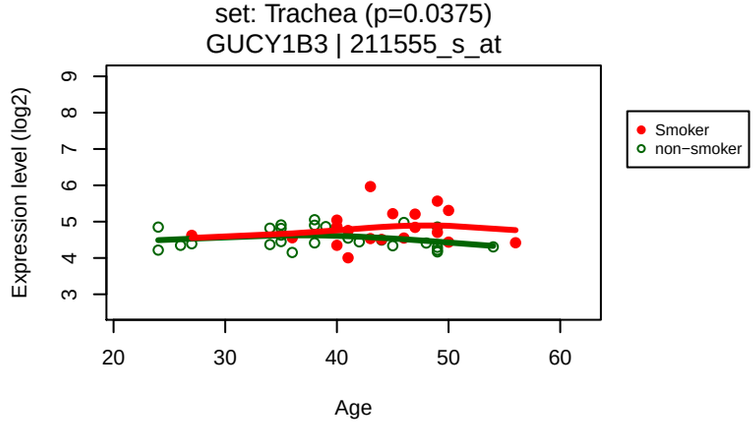

Gene expression ratio in Males
gene: GUCY1B3 | 211555_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5368
p=0.8343
p=0.2307
p=0.0266
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GUCY1B3 | 211555_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9558
p=0.4252
p=0.3115
p=0.0517
SmallAirway2
LargeAirway2
Trachea
Macrophage2

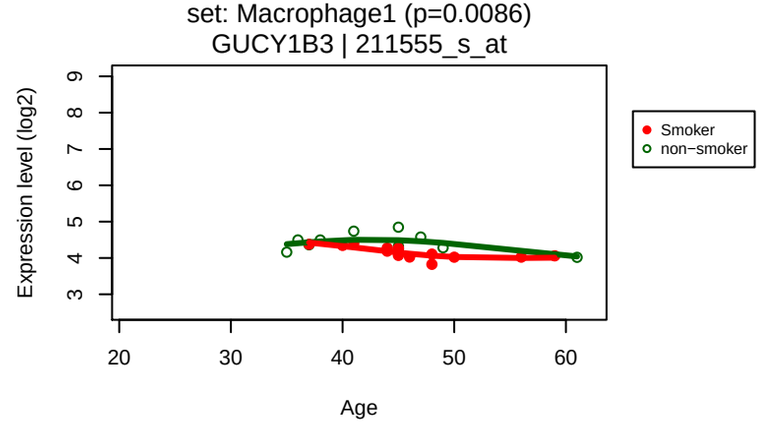

set: SmallAirway1 (p=0.9576)
GUCY1B3 | 211555_s_at
Expression level (log2)
Age
Smoker
non−smoker

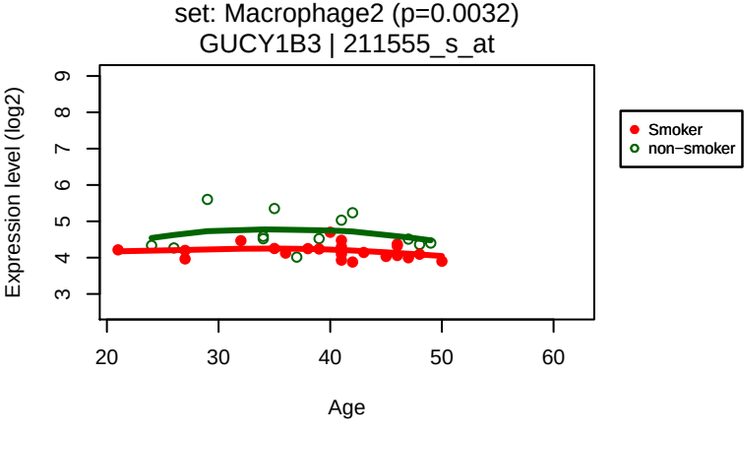

set: SmallAirway2 (p=0.5048)
GUCY1B3 | 211555_s_at
Expression level (log2)
Age
Smoker
non−smoker

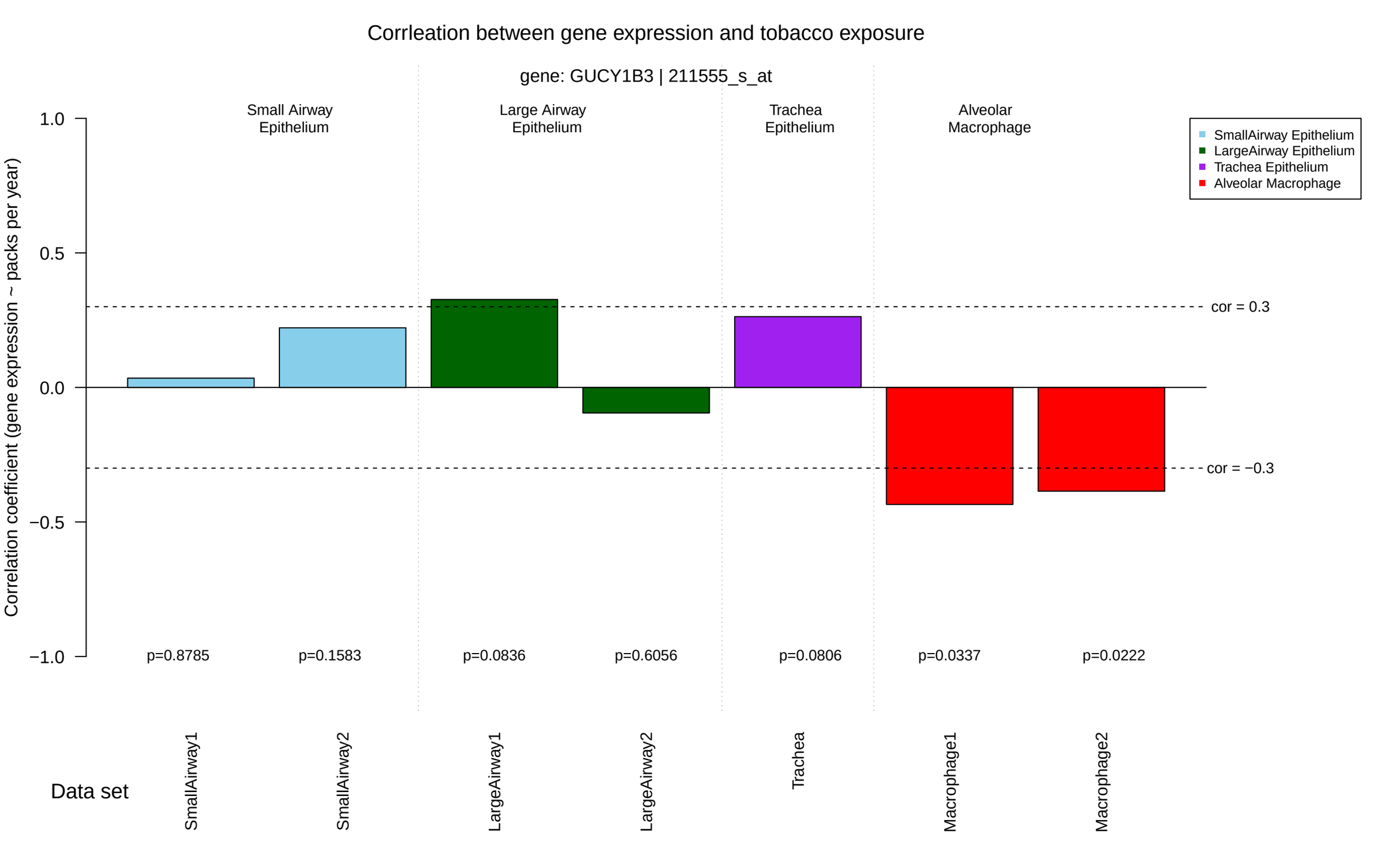

set: LargeAirway1 (p=0.2865)
GUCY1B3 | 211555_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4527)
GUCY1B3 | 211555_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0375)
GUCY1B3 | 211555_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0086)
GUCY1B3 | 211555_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0032)
GUCY1B3 | 211555_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GUCY1B3 | 211555_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8785
p=0.1583
p=0.0836
p=0.6056
p=0.0806
p=0.0337
p=0.0222
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2